Total No. of Questions : 8]

SEAT No. :

PA-1447

[Total No. of Pages : 2

[5926]-63
T.E. (Computer Engineering)
DISTRIBUTED SYSTEMS
(2019 Pattern) (Semester - I) (Elective - I) (310245 C)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagram must be drawn wherever necessary.***
3) ***Assume suitable data, if necessary.***

Q1) a) Describe the importance of clock synchronization algorithm. Explain lamport's logical clock for clock synchronization. **[9]**

b) What is mutual exclusion? Explain centralized algorithm in detail with example. **[9]**

OR

Q2) a) Explain clock synchronization algorithms in detail. **[9]**

b) Describe Gossip-based contribution in detail. **[9]**

Q3) a) Describe the following in brief **[9]**

i) Flat naming

ii) Structured naming

iii) Attributed based naming

b) Explain file service architecture in distributed system. **[8]**

OR

Q4) a) Describe suns network file system in detail. **[9]**

b) Why naming is significance in distributed system? Describe any two types of naming. **[8]**

P.T.O.

Q5) a) Describe consistency protocols in brief. **[9]**

b) What is replica management? Explain techniques of replica management. **[9]**

OR

Q6) a) Describe Cache coherence protocols in detail. **[9]**

b) Describe Data - centric consistency models in detail. **[9]**

Q7) a) What is process resilience? Describe how process resilience can be achieved. **[9]**

b) Explain how reliable client server communication can be achieved. **[8]**

OR

Q8) a) Describe recovery techniques in distributed system. **[9]**

b) Explain how consensus achieved in faulty systems. **[8]**